

PROJECT DESIGN PHASE Solution Design & Prototype Flow

AnswerBot – AI-Powered FAQ & Customer Support Automation

Team Details	
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Team Size	4
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Design Objective

Create a simple backend-driven workflow that allows authorized users to authenticate, search existing FAQs, generate new FAQs using Gemini AI, manage FAQ content and organize information into categories.

System Flow

User → Login/Register → JWT Authentication → FAQ Dashboard → Search Existing FAQ → Result Found? → Review & Respond / Generate FAQ with Gemini → Review/Edit → Save to MongoDB → Reuse FAQ

Core Modules

Authentication

Register, login, JWT token validation and secure password hashing using bcrypt.

FAQ Dashboard

View, create, edit, delete and organize FAQs.

FAQ Search

Search FAQs using keywords and semantic/similarity-based matching.

AI FAQ Generation

Send a question or topic to Gemini AI and receive a draft FAQ.

Category Management

Group FAQs into meaningful categories.

Role Management

Provide appropriate permissions to administrators, content creators and authenticated users.

Validation & Errors

Validate requests and provide consistent error responses.

Technology Architecture

Client → REST API → Node.js + Express.js → Authentication & Services → MongoDB + Mongoose

AI Integration: Node.js/Express service → Google Gemini AI → Generated FAQ → User Review → MongoDB

Prototype Success Criteria

- User can register and log in securely.
- Authorized users can search existing FAQs.

- AI can generate a draft FAQ.
- Users can review and modify generated content.
- FAQs can be created, updated and deleted.
- FAQs can be organized by category.
- Private functions are protected through authentication and authorization.

Design Principle

Keep the workflow simple: **Search first** → **Generate when needed** → **Review** → **Save** → **Reuse**. AI assists the support team while human users retain control over the final content.